# LAB SHEET–02

# Solved Questions and Answers

BCA 3rd Semester | Session 2026–2027

Web Technology Lab — Short Codes with Visual Output Examples

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# Q1. Student Registration Form

**Question:** Create a student registration form containing name, email, gender, course, hobby, address, Submit and Reset controls.

### Answer / Source code — q1.html

```
<!DOCTYPE html>
<html>
<head>
<style>
body { font: 14px Arial; background: #eee; }
form { background: white; padding: 10px; width: 300px; }
input, select, textarea { margin: 3px; }
</style>
</head>
<body>
<h3>Student Registration</h3>
<form>
Name: <input type="text"><br>
Email: <input type="email"><br>
Gender: <input type="radio" name="g">M
        <input type="radio" name="g">F<br>
Course: <select><option>BCA</option><option>BBA</option></select><br>
Hobby: <input type="checkbox">Reading<br>
Address: <textarea rows="2"></textarea><br>
<input type="submit"><input type="reset">
</form>
</body>
</html>
```

### Expected output:

```
Student Registration
Name:       [_________________]
Email:      [_________________]
Gender:     ( ) M  ( ) F
Course:     [ BCA v ]
Hobby:      [ ] Reading
Address:    [_________________]
            [ Submit ] [ Reset ]
```

**Result:** The form displays text and email inputs, radio buttons, a checkbox, a dropdown, a textarea and form buttons.

# Q2. College Website Using Frames

**Question:** Create a college website using frames. Divide the page into a header, navigation menu and main content area. Menu links should open in the main frame.

### Answer / Source code — save these files in the same folder

#### q2.html

```
<!DOCTYPE html>
<html>
<frameset rows="20%,80%">
  <frame src="header.html">
  <frameset cols="25%,75%">
    <frame src="menu.html">
    <frame src="home.html" name="main">
  </frameset>
</frameset>
</html>
```

#### header.html

```
<h2 align="center">COER University</h2>
```

#### menu.html

```
<a href="home.html" target="main">Home</a><br>
<a href="courses.html" target="main">Courses</a><br>
<a href="about.html" target="main">About</a>
```

#### home.html

```
<h3>Welcome to COER University</h3>
<p>Main content appears here.</p>
```

#### courses.html

```
<h3>Courses: BCA, BBA, MCA</h3>
```

#### about.html

```
<h3>About Our College</h3>
```

### Expected output:

```
COER University
+----------------+--------------------------------------+
| Menu           | Welcome to COER University           |
| Home           | Main content appears here.           |
| Courses        |                                      |
| About          |                                      |
+----------------+--------------------------------------+
```

**Result:** The page is divided into a header, navigation menu and main frame. Clicking Home, Courses or About loads that page in the main frame. Traditional frames are obsolete in HTML5 but match the lab sheet.

# Q3. Login and Admission Form

**Question:** Create a login form with email and password fields, and an admission form with name, email and course fields.

### Answer / Source code — q3.html

```
<!DOCTYPE html>
<html>
<head>
<style>
body { font: 14px Arial; background: #eee; }
form { background: white; padding: 8px; width: 280px; margin-bottom: 10px; }
input, select { margin: 3px; }
button { background: blue; color: white; }
</style>
</head>
<body>
<h3>Login</h3>
<form>
Email: <input type="email"><br>
Password: <input type="password"><br>
<button>Login</button>
</form>
<h3>College Admission</h3>
<form>
Name: <input type="text"><br>
Email: <input type="email"><br>
Course: <select><option>BCA</option><option>BBA</option></select><br>
<button>Apply</button>
</form>
</body>
</html>
```

### Expected output:

```
Login
Email:       [__________________]
Password:    [__________________]
             [ Login ]

College Admission
Name:        [__________________]
Email:       [__________________]
Course:      [ BCA v ]
             [ Apply ]
```

**Result:** Both forms are displayed with basic CSS formatting.

# Q4. JavaScript Calculator

**Question:** Create a calculator that performs addition, subtraction, multiplication and division on two numbers. Handle division by zero.

### Answer / Source code — q4.html

```
<!DOCTYPE html>
<html>
<body>
<h3>Calculator</h3>
<input id="a" type="number" placeholder="First number">
<input id="b" type="number" placeholder="Second number"><br>
<button onclick="c('+')">+</button>
<button onclick="c('-')">-</button>
<button onclick="c('*')">*</button>
<button onclick="c('/')">/</button>
<p id="r">Result:</p>
<script>
function c(o) {
  let a = +document.getElementById('a').value;
  let b = +document.getElementById('b').value;
  let r;
  if (o == '+') r = a + b;
```

```
  else if (o == '-') r = a - b;
  else if (o == '*') r = a * b;
  else r = b == 0 ? 'Cannot divide by zero' : a / b;
  document.getElementById('r').innerHTML = 'Result: ' + r;
}
</script>
</body>
</html>
```

### Expected output:

```
Calculator
[ 10 ] [ 5 ]
[ + ] [ - ] [ * ] [ / ]
Result: 15

Example: 10 + 5 = 15
Division by zero: Cannot divide by zero
```

**Result:** The calculator performs four arithmetic operations and checks for division by zero.

## Q5. JavaScript Form Validation

**Question:** Create a registration form and validate required fields, a 10-digit mobile number, password length of at least 6 characters and age from 16 to 100.

### Answer / Source code — q5.html

```
<!DOCTYPE html>
<html>
<body>
<h3>Registration</h3>
<form onsubmit="return v()">
Name: <input id="n"><br>
Email: <input id="e" type="email"><br>
Mobile: <input id="m"><br>
Password: <input id="p" type="password"><br>
Age: <input id="a" type="number"><br>
<button>Register</button>
</form>
<p id="msg"></p>
<script>
function v() {
  let g = id => document.getElementById(id).value.trim();
  let n = g('n'), e = g('e'), m = g('m'), p = g('p');
  let a = +g('a'), s = document.getElementById('msg');
  if (!n || !e || !m || !p || !g('a'))
    s.innerHTML = 'All fields required';
  else if (!/^\\d{10}$/.test(m))
    s.innerHTML = 'Enter 10 digit mobile';
  else if (p.length < 6)
    s.innerHTML = 'Password needs 6 characters';
  else if (a < 16 || a > 100)
    s.innerHTML = 'Age must be 16-100';
  else
    s.innerHTML = 'Registration successful!';
  return false;
}
</script>
</body>
</html>
```

### Expected output:

```
Student Registration
Name:        [__________________]
Email:       [__________________]
Mobile:      [__________________]
Password:    [__________________]
Age:         [__________________]
             [ Register ]

Example: Enter 10 digit mobile
```

```
Valid entries: Registration successful!
```

**Result:** JavaScript checks empty fields, mobile number, password length and age, then displays a validation message.

## How to Run the Programs

1. Open Notepad or VS Code.
2. Copy each program into the file with the exact filename shown and save it with the .html extension.
3. Open the HTML file in a web browser.
4. For Q2, save q2.html, header.html, menu.html, home.html, courses.html and about.html together in the same folder, then open q2.html.

**Note:** Output sketches describe the expected browser display. Exact appearance may vary by browser and operating system.